RITHIKA BABU

Software Engineer

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PROFESSIONAL SUMMARY

Software Engineer with 1.8+ years of relevant, hands-on experience across the full software development lifecycle — design, development, testing, and deployment — applying solid software engineering principles with minimal supervision. Experienced building and maintaining production applications using React, TypeScript, Node.js, and NestJS, with working knowledge of REST/GraphQL API design, PostgreSQL schema design, and cloud deployment (AWS EC2, RDS, S3). Works co-operatively within Agile/Scrum teams, collaborating closely with product managers and fellow engineers to deliver reliable, accurate results, and has consistently made sound judgment calls based on situational understanding rather than rote process. Progressed from intern to full-time engineer within one quarter through consistent, dependable delivery. Currently pursuing an MCA specializing in AI/ML.

TECHNICAL SKILLS

Programming & Software Design: JavaScript (ES6+), TypeScript, Python, OOP concepts, software design principles, system design

Frontend: React.js, Redux Toolkit, React Query, HTML5, CSS3, Tailwind CSS, Material UI, Storybook

Backend: Node.js, NestJS, Express.js, REST APIs, GraphQL, Prisma ORM, JWT, OAuth 2.0

Databases: PostgreSQL, MySQL, Elasticsearch (NoSQL), schema design, database modeling, query optimization

DevOps & Cloud: Docker, Kubernetes (AKS), AWS (EC2, RDS, S3), CI/CD, GitHub Actions

Testing: Jest, Supertest, Playwright (E2E), Unit & Integration Testing

Tools & Practices: Git, GitHub, Postman, Swagger, Jira, Agile/Scrum, code reviews

WORK EXPERIENCE

Full Stack Developer — CreamCollar

Oct 2024 – Present

Full-Time (progressed from Intern within first quarter)

- Applied software engineering principles to design and build scalable REST and GraphQL APIs using NestJS, TypeScript, and Prisma ORM, working independently with minimal supervision to deliver accurate, reliable results.
- Led frontend architecture for a storefront migration to a modern React + TypeScript stack, exercising original judgment in technical decisions and collaborating closely with product managers and backend engineers.
- Built performant, accessible storefront UI — product listing, cart, checkout, order management — with full cross-browser compatibility.
- Implemented JWT and OAuth 2.0-based authentication, role-based access control (RBAC), and third-party SSO integration across backend services, ensuring secure and reliable system behavior.
- Built dynamic, data-driven dashboards and admin panels using Material UI and Tailwind CSS, translating stakeholder requirements into working software.
- Integrated LLM-powered features via Azure OpenAI and external AI APIs using prompt engineering, enabling intelligent recommendations and automated content generation.
- Implemented E2E and unit/integration test automation (Playwright, Jest, Supertest), consistently producing accurate, well-tested code across critical user flows.
- Optimized frontend performance through code splitting, lazy loading, memoization, and Core Web Vitals tuning, improving load speed and overall user experience.
- Deployed and maintained application infrastructure on AWS (EC2, RDS, S3); configured Docker containers and CI/CD pipelines via GitHub Actions, taking ownership of the release process.
- Designed and maintained PostgreSQL schemas, migrations, and indexing strategies, applying sound database design judgment to support evolving business requirements.
- Worked co-operatively in Agile/Scrum ceremonies (sprint planning, standups, retrospectives), directly and positively impacting sprint delivery and team performance.
- Managed source control using Git/GitHub with feature-branch workflows, participating actively in code reviews and architecture discussions.

PROJECTS

Skill Gap Analysis — Full Stack Application
React, TypeScript, NestJS, Azure OpenAI, REST API

- Built a full-stack application with a React + TypeScript frontend and NestJS backend, using Azure OpenAI to power personalized skill recommendations via a modular, scalable architecture.

Barrier Cord Sensor — AI-Powered Emergency Detection System
Python, Machine Learning, Hardware Integration

- Developed a real-time AI emergency sensor system in Python with ML models for detection accuracy and automated hardware-triggered alert mechanisms.

Blood Group Detection Using Fingerprint — Full Stack Application
Python, React, NestJS, Machine Learning

- Engineered a biometric prediction system with a Python ML backend and React + NestJS full-stack architecture in a scalable, modular client-server design.

EDUCATION

Master of Computer Applications (MCA) — AI/ML Specialization *Sep 2025 – Apr 2027*
SRM University

Bachelor of Computer Applications (BCA) — Data Science Specialization *Aug 2022 – Apr 2025*
Vels Institute of Science, Technology & Advanced Studies (VISTAS)